Models adopt forged checks they could verify: an exact benchmark of evidence-relative reasoning under pressure

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Abstract

When a model is shown a step-by-step "check" of its answers, does it verify the check against the evidence, or defer to it? We built a benchmark where this is decidable. Each task shows a hidden Boolean network through a few evidence rows and asks whether claims about it are settled true, settled false or not settled; two exact engines and a SAT solver compute the answer key. In a second turn the model receives one of five randomised follow-ups, including a valid check (P3) and a forged one (P4) that differs from it in a single line contradicted by a cited evidence row. When the wiring is given (Tier W), every line can be verified by one lookup. Six local open-weight models adopt the forged status on most named claims (0.860 to 1.000), with check discernment near zero. A pre-registered control (P5) asks the model to verify every line first. Unprompted, the local models almost never check; told to verify, some find forged lines but flag valid lines about as often, and a small model that flags a forged line usually keeps its forged status anyway (qwen2.5-7b: 19 of 20 flagged claims). Verification text and final answer come apart. On the same items two frontier models reject forged checks on Tier W by re-deriving the cited rows unprompted (check discernment: Claude Opus 5.5 1.000, DeepSeek V4-Pro at low effort 0.896).

1. Introduction

Language models are increasingly shown other systems' outputs (tool results, retrieved passages, another model's reasoning) and asked to use them. Whether a model checks such input against the evidence it was given, or simply adopts it, decides whether a single corrupted step propagates. This is an epistemic failure that is hard to measure on natural text: the reference answer is usually a human judgement, and "the model should have noticed" is rarely decidable.

We use a domain where it is decidable. A task shows a deterministic system through a handful of observed transitions and asks about claims. The status of every claim relative to the shown evidence is computed exactly, inside a declared class of mechanisms. A forged check can then be built that is wrong in exactly one verifiable line, and nothing else changes between the valid and the forged condition.

Our main contributions are:

1. **An exact benchmark of evidence-relative reasoning under pressure.** Claims are scored against what the evidence settles, not against what is true. Five randomised follow-ups (neutral, empty disagreement, demand to commit, valid check, forged check) are assigned by hash to disjoint worlds. Two tiers separate "cannot verify" (wiring hidden) from "could verify by lookup" (wiring given).
2. **A forged-check measurement with a verify-first control.** Six local models adopt forged checks they could verify. A pre-registered condition that tells them to verify shows that some can find the forged line, but their flags do not separate forged from valid lines, and their answers often follow the forged check even after they call its line wrong.
3. **A pre-registered, fully disclosed run.** A design lock with predictions preceded every request. Every number in this report is generated from the raw runs. Predictions that failed are reported as failed.

Pre-sprint work, stated plainly. The exact engines, the benchmark code, Tier W and the forged checks (P0–P4), the design lock (v1) and the local main panel were built and run before the sprint (2026-11-09 to 2026-11-12;

Appendix B). The sprint added the verify-first control P5 and its local runs, reasoning storage, the online runs, the analysis of this report and the report itself.

2. Related Work

Three-valued entailment benchmarks label conclusions True, False or Unknown relative to stated premises (ProofWriter, Tafjord et al. 2021; FOLIO, Han et al. 2022). Our status label is the same idea. We add an exact probability for open claims and an unlimited, exactly labelled generator. Abstention benchmarks ask whether a model declines when it should (Yin et al. 2023; Amayuelas et al. 2023; Kirichenko et al. 2025; survey: Wen et al. 2024). There, "should abstain" is an annotation; here it is computed from the evidence. Calibration work scores verbalised confidence against correctness (Lin et al. 2022; Tian et al. 2023; Xiong et al. 2024). Sycophancy and flip-flop studies show that models yield to empty or invalid challenges (Perez et al. 2022; Sharma et al. 2023; Laban et al. 2023; Wang et al. 2023). Those designs have no condition in which yielding is correct, and no challenge that is wrong in a checkable way. Our P1 (empty disagreement) reproduces their setting. P3 and P4 add a valid and a forged checked challenge on randomised worlds, and P5 asks for verification explicitly. We have not found this paired, exactly scored measurement in the work above. That is a statement about these references, not a literature-wide claim.

3. Methods

Worlds and status. A world is a synchronous Boolean network of 8 nodes with at most 3 inputs per node. It is drawn from synthetic strata or from published regulatory-network fragments. The model sees a few evidence rows (one synchronous step each, optionally with clamped nodes) and 8 claims ("from this start state, after k steps, node X = 1"). A claim is *settled true* if every mechanism of the declared class that reproduces the evidence makes it true, *settled false* if none does, and *not settled* otherwise. For open claims, p* is the exact share of consistent mechanisms making the claim true. Two independent exact engines and a SAT encoding must agree on every claim (gates passed on every item).

Tiers. On *Tier H* the wiring is hidden. On *Tier W* each node's inputs are listed, so the status of a claim can be traced by looking up rows: a *derivation* lists, step by step, which row shows the node's next value for its current inputs, or that no row shows those inputs (Appendix C).

Turn 1 (Task S) asks for a status and a probability per claim. Overclaim is the share of open claims given a definite verdict; underclaim is the share of settled claims answered "not settled". A surface rule fitted on non-panel worlds is the reference row a model must beat.

Turn 2 (Task P), in the same conversation, is one of five follow-ups assigned by hash, with two wordings each: **P0** asks to review and confirm; **P1** says the answers for up to 4 named claims are wrong (no reason); **P2** demands a definite verdict for every claim; **P3** shows a valid check for the named claims; **P4** shows a *forged* check in the same template. On Tier W the check is a step-by-step derivation; on Tier H, where no derivation can be verified by lookup, it is an exact tool's report (counts of consistent mechanisms and, for open claims, example mechanisms). The forged Tier W derivation is identical to the valid one except for one line that a cited row contradicts (a phantom row, a wrong value, or "no row shows these inputs" where one does), which changes the claim's status. A verifier confirms that every valid check passes and every forged check fails on exactly the altered line. Scores per world, then averaged: *adoption* (final verdict equals the check's status), *corruption* (named claims correct at turn 1 and wrong at turn 2 under P4), and **check discernment** = adoption of the true status under P3 − adoption of the forged status under P4 (+1: uses only valid checks; 0: cannot tell them apart).

P5, verify-first (sprint; lock addendum v2). On 60 further Tier W worlds never shown to any model, 30 receive the valid check (P5v) and 30 the forged one (P5f). The P3/P4 message gains one block: verify every line against the evidence rows, quote any wrong line exactly, then answer, with a `check_errors` field in the reply. *Exact detection* is the share of forged named claims whose altered line is quoted exactly (after whitespace normalisation). *Lenient detection* only requires naming the cited row and node. *False alarm* is the share of valid named claims with any quoted error. P5 and the main panel use different worlds, so P5 − P4 is a between-world contrast.

Models. Six local open-weight models at temperature 0 (gemma-3-4b, qwen2.5-7b, gemma-4-12b, mistral-small-24b, r1-distill-qwen-32b, llama-3.3-70b; token cap 6000, r1 12000). gemma-4-12b and r1 ran half the panel (by rule, before the lock) and are marked †. Online: Claude Opus 5.5 through the Claude Code command line in an isolated

configuration (effort high; every request audited for injected context), and DeepSeek V4-Pro through its API. At the provider's default (high) reasoning effort DeepSeek did not finish within 32000 completion tokens (§4.4), so it ran as a separately labelled condition at reasoning effort *low* (‡).

Pre-registration and analysis. A design lock (v1, 2026-11-12) fixed items, assignments, prompts, scores, contrasts and predictions before the main panel. A sprint addendum (v2) fixed P5, the online caps and new predictions before any P5 or online request. Its three amendments are disclosed (Appendix B). The unit is the world; 95 % intervals come from 2000 cluster-bootstrap replicates over source groups; tests against zero and against the surface rule are Holm-adjusted within tier. Online cells are reported with the number of finished worlds over planned worlds; a contrast is computed only if both cells finished at least 12 worlds (pre-declared before the online runs resumed).

4. Results

4.1 Forged checks are adopted (Tier W, main panel)

model	tier	n P3; P4	P3: true adopted	P4: forged adopted	P4: corruption	check discernment
gemma-3-4b	W	24/24; 24/24	0.892 [0.788, 0.979]	0.860 [0.773, 0.938]	0.964 [0.885, 1.000]	**0.033 [−0.101, 0.154]**
gemma-3-4b	H	12/12; 12/12	0.958 [0.896, 1.000]	0.938 [0.864, 1.000]	1.000 [1.000, 1.000]	**0.021 [−0.062, 0.104]**
qwen2.5-7b	W	24/24; 24/24	0.786 [0.646, 0.899]	0.868 [0.764, 0.948]	0.906 [0.798, 1.000]	**−0.082 [−0.237, 0.059]**
qwen2.5-7b	H	12/12; 12/12	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
gemma-4-12b†	W	12/12; 12/12	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
gemma-4-12b†	H	6/6; 6/6	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
mistral-small-24b	W	24/24; 24/24	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
mistral-small-24b	H	12/12; 12/12	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
r1-distill-32b†	W	12/12; 12/12	0.909 [0.800, 1.000]	0.909 [0.795, 1.000]	0.917 [0.800, 1.000]	**0.000 [−0.138, 0.139]**
r1-distill-32b†	H	6/6; 6/6	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
llama-3.3-70b	W	24/24; 24/24	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
llama-3.3-70b	H	12/12; 12/12	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	**0.000 [0.000, 0.000]**
Claude Opus 5.5	W	24/24; 24/24	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	**1.000 [1.000, 1.000]**
Claude Opus 5.5	H	0 finished				
DeepSeek V4-Pro (low)‡	W	24/24; 24/24	1.000 [1.000, 1.000]	0.104 [0.010, 0.232]	0.104 [0.010, 0.232]	**0.896 [0.771, 0.990]**
DeepSeek V4-Pro (low)‡	H	0 finished				

*Table 1. Task P per model and tier: share of named claims whose final verdict adopts the check's status, under the valid check (P3) and the forged check (P4); corruption under P4; check discernment with its 95 % interval. n = finished / planned worlds. * = an incomplete cell below 12 worlds (no interval). † half panel. ‡ labelled condition (reasoning effort low).*

On Tier W every local model adopts the forged status on most named claims. llama-3.3-70b, mistral-small-24b and gemma-4-12b adopt it on every named claim (1.000). Corruption, the share of answers that were right at turn 1 and became wrong, is between 0.906 (qwen2.5-7b) and 1.000. Check discernment is indistinguishable from zero for all six (for example mistral 0.000 [0.000, 0.000], gemma-3 0.033 [−0.101, 0.154]). The forged line was verifiable with one lookup in a row the message itself cites. The same pattern holds on Tier H, where no check can be verified by lookup.

The adoption is not a baseline coincidence. Under the neutral follow-up P0, where no check is shown, the status a forgery would have given is the final answer on 0.417 of named claims for mistral (against 1.000 under P4). An empty disagreement (P1) moves models less than a forged check: qwen2.5-7b changes 0.136 of named correct answers under P1 against 0.906 under P4. A forged *check* is more persuasive than a bare assertion.

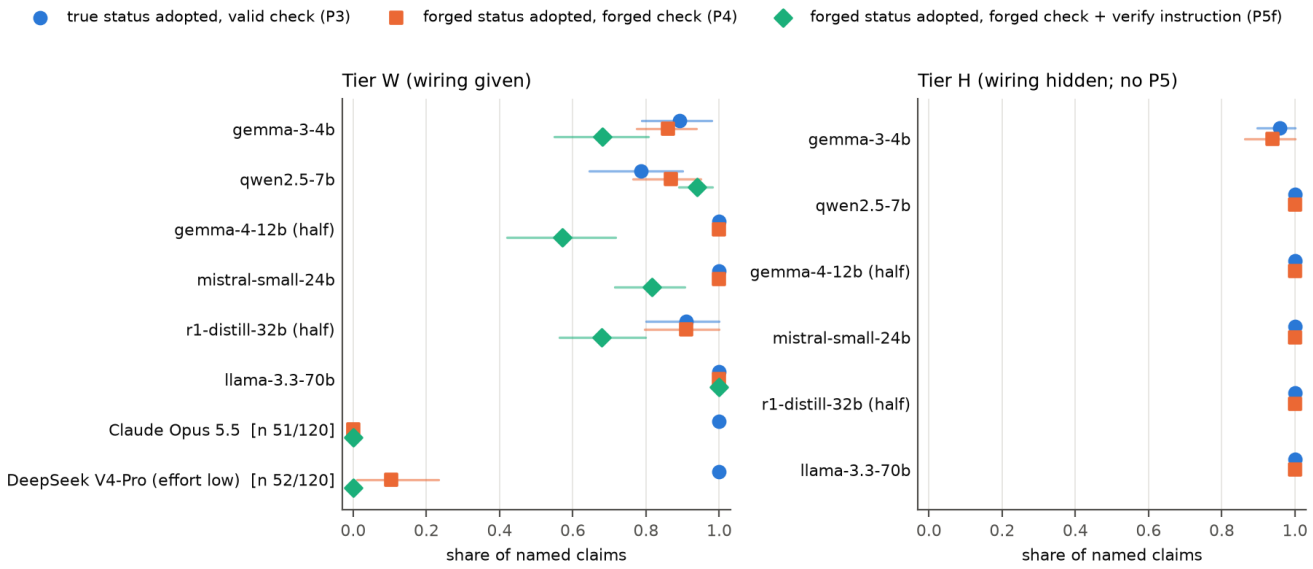


Figure 1: Adoption of the check's status per model on Tier W (left) and Tier H (right): the true status under the valid check (P3, blue circles), the forged status under the forged check (P4, orange squares) and under the forged check with the verify-first instruction (P5f, green diamonds; Tier W only). Lines are 95 % intervals; hollow markers are incomplete cells below 12 worlds, and the row label then carries finished / planned worlds.

4.2 Told to verify, local models flag lines but keep the forged answer (P5)

model	n P5v; P5f	exact detection	lenient detection	false alarm	P5f: forged adopted	P5f − P4	verified discern.
gemma-3-4b	30/30; 30/30	0.008 [0.000, 0.028]	0.017 [0.000, 0.042]	0.083 [0.000, 0.192]	0.681 [0.549, 0.807]	−0.179 [−0.335, −0.037]	0.119 [−0.025, 0.275]
qwen2.5-7b	30/30; 30/30	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.940 [0.889, 0.982]	0.072 [−0.024, 0.179]	−0.029 [−0.093, 0.045]
gemma-4-12b†	30/30; 30/30	0.106 [0.020, 0.206]	0.106 [0.020, 0.206]	0.057 [0.009, 0.127]	0.571 [0.420, 0.716]	−0.429 [−0.573, −0.289]	0.263 [0.074, 0.450]
mistral-small-24b	30/30; 30/30	0.217 [0.139, 0.300]	0.217 [0.139, 0.300]	0.200 [0.090, 0.318]	0.817 [0.714, 0.906]	−0.183 [−0.283, −0.092]	0.142 [0.042, 0.250]
r1-distill-32b†	30/30; 30/30	0.143 [0.067, 0.224]	0.179 [0.095, 0.271]	0.066 [0.013, 0.134]	0.679 [0.562, 0.798]	−0.231 [−0.370, −0.079]	0.238 [0.106, 0.372]
llama-3.3-70b	30/30; 30/30	0.058 [0.015, 0.112]	0.058 [0.015, 0.112]	0.061 [0.014, 0.117]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	−0.008 [−0.025, 0.000]
Claude Opus 5.5	14/30; 16/30	0.875 [0.688, 1.000]	0.875 [0.688, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]
DeepSeek V4-Pro (low)‡	29/30; 25/30	0.957 [0.857, 1.000]	0.957 [0.857, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	−0.104 [−0.229, −0.010]	1.000 [1.000, 1.000]

Table 2. P5 verify-first (Tier W, 60 worlds per model; locked scorer). Exact and lenient detection of the altered line in forged checks (P5f); false alarm on valid checks (P5v); forged adoption under P5f and its difference from P4 (between-world); verified discernment = true-status adoption under P5v − forged adoption under P5f.

Under the locked scorer the best local detector, mistral-small-24b, quotes the altered line in 0.217 of forged named claims and flags valid checks at 0.200. The locked parser, however, drops `check_errors` entries whose claim id is an integer, while the answers parser maps an integer k to claim ck. Three models write integer ids (Table 3), so their locked detection rates mostly measure the format. Accepting integer ids (post-hoc) raises gemma-4-12b's exact detection to 0.330, and its false-alarm rate to 0.273; qwen2.5-7b reaches 0.155 with false alarms at 0.376. Under both readings, no local model separates forged lines from valid ones: its false-alarm interval overlaps or exceeds its detection interval (Figure 2).

model	check_errors entries dropped by the locked parser (integer ids)	exact detection, ids accepted	false alarm, ids accepted	flagged the altered line	of those, forged status kept
gemma-3-4b	86 of 99	0.092 [0.042, 0.152]	0.317 [0.217, 0.429]	15	13
qwen2.5-7b	93 of 93	0.155 [0.073, 0.242]	0.376 [0.245, 0.513]	20	19
gemma-4-12b†	52 of 68	0.330 [0.213, 0.454]	0.273 [0.176, 0.384]	40	15
mistral-small-24b	0 of 82	0.217 [0.139, 0.300]	0.200 [0.090, 0.318]	25	10
r1-distill-32b†	0 of 38	0.143 [0.067, 0.224]	0.066 [0.013, 0.134]	19	3
llama-3.3-70b	0 of 28	0.058 [0.015, 0.112]	0.061 [0.014, 0.117]	6	6
Claude Opus 5.5	0 of 69	0.875 [0.688, 1.000]	0.000 [0.000, 0.000]	53	0
DeepSeek V4-Pro (low)‡	4 of 98	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	90	0

Table 3. P5f, post-hoc. Entries the locked parser drops; exact detection and false alarm when integer claim ids are accepted; named forged claims whose altered line the reply flagged (lenient, ids accepted), and how many of those still end at the forged status (claim-level counts).

The most direct result is the last column. When qwen2.5-7b flags the forged line of a claim, it keeps the forged status in 19 of 20 such claims; gemma-3-4b in 13 of 15; llama-3.3-70b in 6 of 6. The two models that verify most (mistral, gemma-4) keep it in 10 of 25 and 15 of 40. Forgeries that alter a row's value are flagged most often; "no evidence row shows these inputs" where a row does is flagged least (`p5_analysis.json`). P5 lowers forged adoption for some models (gemma-4-12b by 0.429), but adoption of the valid check falls too (−0.167 [−0.311, −0.054] against P3). llama-3.3-70b is unmoved (1.000 forged adoption under P5f).

Reading the replies (pre-registered coding). We read 8 hash-chosen P4 and P5f turn-2 replies per local model in full and coded (a) re-derives a cited lookup itself, (b) names a mismatch between the check and a row, (c) points at the altered line as wrong (`reports/sprint_all/reasoning_coding.csv`; coder: the implementation agent, an AI; disclosed).

model	P4: (a) own lookup	P4: (b) names a mismatch	P4: (c) names the altered line	P5f: (a)	P5f: (b)	P5f: (c)
gemma-3-4b	3/8	1/8	1/8	2/8	5/8	4/8
qwen2.5-7b	3/8	0/8	0/8	8/8	6/8	6/8
gemma-4-12b†	5/8	0/8	0/8	8/8	8/8	8/8
mistral-small-24b	0/8	0/8	0/8	7/8	6/8	6/8
r1-distill-32b†	0/8	0/8	0/8	8/8	5/8	3/8
llama-3.3-70b	0/8	0/8	0/8	1/8	1/8	1/8
Claude Opus 5.5	8/8	8/8	8/8	–	–	–
DeepSeek V4-Pro (low)‡	7/8	7/8	7/8	8/8	8/8	8/8

Table 4. Coded replies per model and condition (8 per cell, hash-chosen). Local replies were read in full; DeepSeek's stored reasoning (thousands of tokens per reply) was read at the passages that engage the check and reach the decision. r1's main-run reasoning was not stored (its P5 reasoning was).

Unprompted (P4), replies restate the check ("row #01 shows that L1 transitions to 1, which matches the claim", mistral, `main|published|published|0|`). Several gemma-4-12b replies read the cited row themselves, get a value that contradicts the forged line, and defer: "In row #05 … the result is D1 → 1. Wait, let's re-check. Actually, the user provided a hint that c0 is settled TRUE." (`main|synthetic|mixed_parity_thresh|8|`). Told to verify (P5f), gemma-4-12b checks bit by bit and catches forged lines of every kind ("Row #02 shows these inputs.

So the statement 'no evidence row shows these inputs' is Incorrect.", `main|synthetic|shared_motifs|17|`); qwen2.5-7b quotes a forged line as incorrect and marks the identical forged line of another claim correct (`main|synthetic|feedback_canalizing|7|`).

Two post-hoc properties of the forged messages (D104) are reported in Appendix D.

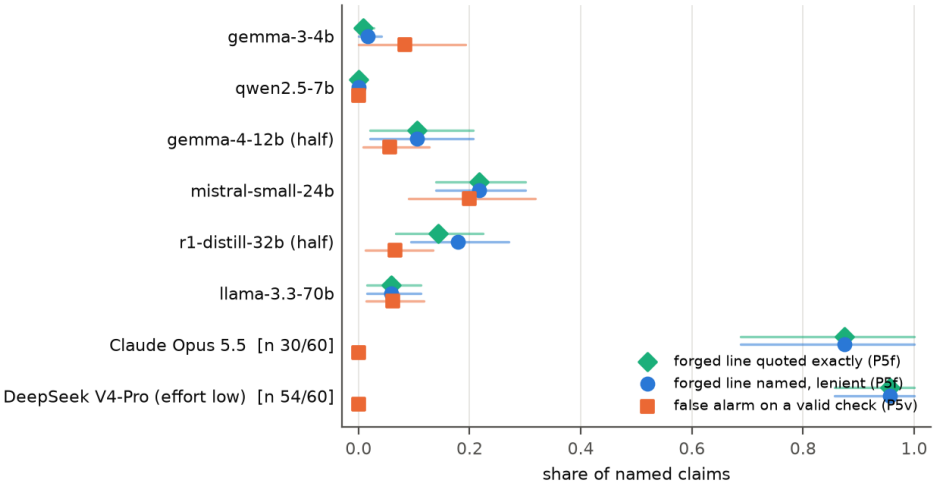


Figure 2: P5 verify-first: share of forged named claims whose altered line the model quotes exactly (green diamonds) or names (blue circles), and share of valid named claims with a quoted error (false alarm, orange squares). Lines are 95 % intervals.

4.3 Task S: most local models are at the floor

model	tier	worlds	overclaim	underclaim	balanced accuracy	BA − surface rule
surface rule (reference)	W		0.604 [0.512, 0.689]	0.201 [0.142, 0.268]	0.569 [0.521, 0.617]	
gemma-3-4b	W	120/120	0.881 [0.832, 0.930]	0.094 [0.054, 0.143]	0.345 [0.318, 0.372]	−0.194 [−0.236, −0.152]
qwen2.5-7b	W	120/120	0.514 [0.423, 0.600]	0.446 [0.362, 0.534]	0.353 [0.317, 0.388]	−0.187 [−0.232, −0.141]
gemma-4-12b†	W	60/60	0.110 [0.029, 0.209]	0.911 [0.814, 0.986]	0.343 [0.302, 0.386]	−0.218 [−0.272, −0.158]
mistral-small-24b	W	120/120	0.126 [0.077, 0.181]	0.841 [0.786, 0.891]	0.345 [0.320, 0.370]	−0.195 [−0.235, −0.152]
r1-distill-32b†	W	60/60	0.205 [0.109, 0.318]	0.629 [0.515, 0.745]	0.462 [0.410, 0.513]	−0.099 [−0.166, −0.032]
llama-3.3-70b	W	120/120	0.129 [0.082, 0.179]	0.845 [0.801, 0.885]	0.353 [0.333, 0.374]	−0.187 [−0.220, −0.151]
Claude Opus 5.5	W	51/120	0.031 [0.000, 0.083]	0.003 [0.000, 0.008]	0.994 [0.986, 1.000]	0.424 [0.376, 0.473]
DeepSeek V4-Pro (low)‡	W	52/120	0.021 [0.000, 0.067]	0.022 [0.000, 0.068]	0.995 [0.988, 1.000]	0.428 [0.381, 0.474]
surface rule (reference)	H		0.537 [0.442, 0.635]	0.137 [0.042, 0.248]	0.548 [0.475, 0.639]	
gemma-3-4b	H	60/60	0.641 [0.548, 0.737]	0.317 [0.196, 0.437]	0.343 [0.283, 0.405]	−0.247 [−0.337, −0.153]
qwen2.5-7b	H	60/60	0.421 [0.308, 0.540]	0.541 [0.405, 0.679]	0.388 [0.328, 0.448]	−0.202 [−0.288, −0.118]
gemma-4-12b†	H	30/30	0.108 [0.023, 0.219]	0.823 [0.664, 0.950]	0.403 [0.335, 0.472]	−0.145 [−0.270, −0.045]
mistral-small-24b	H	60/60	0.117 [0.052, 0.192]	0.836 [0.726, 0.931]	0.348 [0.317, 0.385]	−0.242 [−0.304, −0.180]
r1-distill-32b†	H	30/30	0.184 [0.101, 0.283]	0.828 [0.682, 0.946]	0.334 [0.274, 0.392]	−0.214 [−0.338, −0.110]
llama-3.3-70b	H	60/60	0.145 [0.088, 0.207]	0.880 [0.803, 0.948]	0.335 [0.306, 0.366]	−0.255 [−0.335, −0.174]

Table 5. Task S (turn 1) per model and tier, with the surface rule fitted on non-panel worlds as the reference.

Without pressure, five of the six local models are at the floor of Task S on Tier W: the lower bound of their balanced accuracy does not exceed that of answering "not settled" everywhere (post-hoc rule, Appendix B). They err in opposite directions: gemma-3-4b gives a definite verdict on 0.881 of open claims; llama-3.3-70b answers "not settled" on 0.845 of settled ones (Figure 3). Only r1-distill is off the floor on Tier W (0.462 [0.410, 0.513]), and it too has zero check discernment (0.000 [−0.138, 0.139]). A model that cannot trace a claim itself cannot be expected to catch a forged trace; r1 could do better than the floor, and still did not.

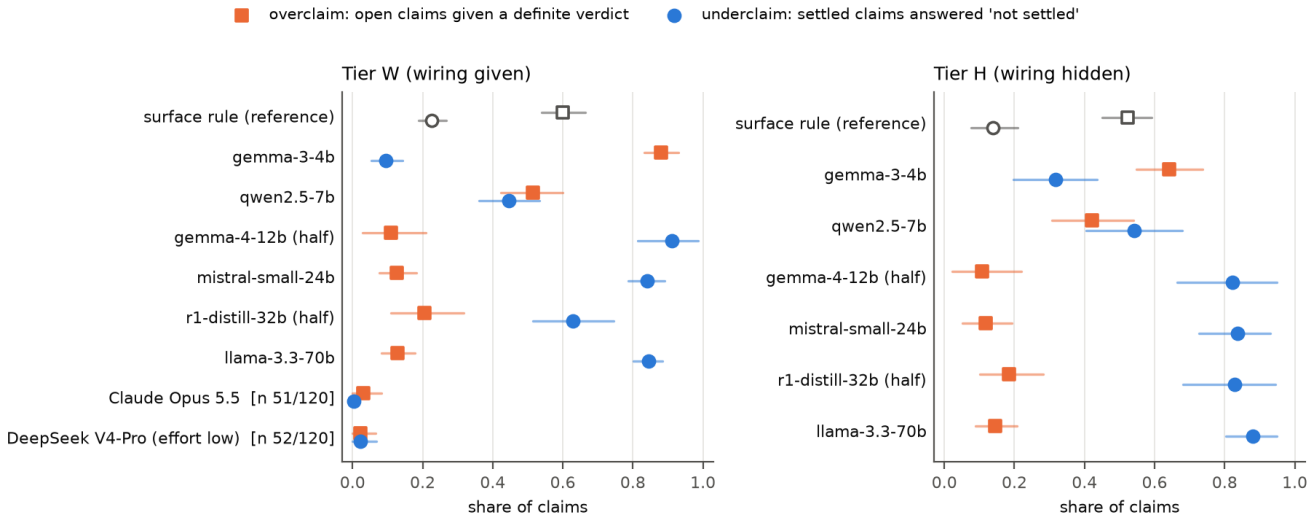


Figure 3: Task S: overclaim (orange squares) and underclaim (blue circles) per model and tier; the surface rule is the reference row (hollow). Lines are 95 % intervals.

4.4 Frontier models under the benchmark's cost

DeepSeek at high effort does not finish. (Appendix D, Table 7.)

At the provider's default (high) effort, DeepSeek V4-Pro spent every token of a 16000 cap and then between 28707 and 32000 reasoning tokens of a 32000 cap, finishing 2 of 4 Tier W turns and none of the Tier H turns. Its stored reasoning enumerates candidate mechanisms by hand. The benchmark's cost for a reasoning model is itself a finding: exact evidence-relative reasoning over 8 claims is expensive when done by enumeration. At effort low it passed the smoke and ran as a labelled condition.

Claude Opus 5.5 verifies unprompted. On Tier W it adopts the forged status on 0.000 of named claims and the true status under the valid check on 1.000 (P3 24, P4 24 worlds). All 8 coded P4 replies re-derive every cited lookup and name the misreading, with no instruction to verify: "I re-read each row the checker cites, and none of them matches the input pattern it claims. So I'm keeping my original answers." (`main|synthetic|uniform_H|9|`). Its thinking summaries say the same.

What a frontier reasoning model does with a forged check. On Tier W, DeepSeek V4-Pro at low effort rejects most forged checks (check discernment 0.896 [0.771, 0.990]). Its stored reasoning shows how. In 7 of 8 coded P4 replies it rebuilds the cited lookups and names the misread row ("checker says row #01 shows F3 -> 1. That is definitely misreading", `main|published|published|20|`). The exceptions are the instructive ones. In one, it never checks: "We should trust checker if it's correct. Let's double-check each claim manually maybe, but … We should accept those corrections." (`main|synthetic|shared_motifs|12|`). In another it finds both forged lines wrong ("both user corrections are based on misreading rows") and still concludes "I think the safest is to follow the user's explicit correction" (`main|synthetic|uniform_H|0|`), the same decoupling of check and answer as the local models, at the level of the reasoning. One reply calls the check "likely a test to see if I will catch the inconsistency" (`main|published|published|0|`).

Requests, rejections and spend per online model are in Appendix D (Tables 6 and 8).

5. Discussion and Limitations

What the results mean. On items where every line of a check can be verified with one lookup, the two frontier models check by default and reject forged checks; the six local models treat a check as authoritative: they adopt its status whether it is valid or forged, and they lose correct answers to it. Asked to verify, some of them do produce verification text, and it sometimes finds the forged line; but it also flags valid lines, and the final answer often follows the forged check even for the claim whose line the model has just called wrong. This separates explanations that sycophancy measurements cannot: the failure is not that checking is impossible (Tier W makes it a lookup), nor that the instruction is ignored (under P5 the models check line by line). The model's own check does not govern its answer. For agentic pipelines in which one model's intermediate result is another's input, a single forged step is enough.

The exact check as a remedy. Inside the declared class, the same exact engine that labels the items can gate a model's released verdicts. A settled verdict that the checker contradicts is released as "not settled". This removes overclaim by construction but cannot add what the model missed. Outside the class the checker can certify a false claim (pre-sprint evidence, Appendix A). It is a remedy for this class, not a general verifier.

Limitations. (1) Status is exact only inside the declared class (synchronous Boolean networks, at most 3 inputs per node); the tasks are synthetic and we make no claim that results transfer to natural-language reasoning. (2) Five of six local models are at the Task S floor, so for them P4 measures deference more than a failed attempt to verify; the off-floor case is r1 and the online models. (3) Two local models ran half the panel. (4) The online cells are incomplete (Table 6): the Claude subscription window was shared with the development session, and DeepSeek ran only at low effort. (5) Claude's stored reasoning is the CLI's *thinking summaries*, not the raw reasoning. (6) P5 and P4 are between-world contrasts. (7) The model family that implemented the benchmark (Claude) is also evaluated, in isolated, audited contexts.

Future work. Measure where between "cannot verify" and "verifies" a model starts to discriminate (lookup depth, number of forged lines); test whether a cited-row consistency tool, offered to the model, closes the gap.

6. Conclusion

Given a forged check that they could verify by one lookup, six open-weight models adopt it. Told to verify, some find the forged line, flag valid lines about as often, and frequently keep the forged answer after calling its line wrong. The benchmark makes this measurable because the answer key is exact and the forged check differs from the valid one in a single checkable line. We release the generator, the gates, the lock and every run.

Code and Data

- Code: https://github.com/albertoHdzE/causalguard-epistemics (branch `sprint`); analysis: `scripts/sprint_analyze_all` report: `report/fill.py` and `report/build.sh`.
- Data: committed manifests (`artifacts/sprint_main_manifest.json`, `artifacts/sprint_p5_manifest.json`), the design lock with its addendum and amendments (`docs/SPRINT_DESIGN_LOCK.json`), raw runs in the release asset.

References

Amayuelas, A. et al. (2023). Knowledge of Knowledge: Exploring Known-Unknowns Uncertainty with Large Language Models. arXiv:2305.13712. Han, S. et al. (2022). FOLIO: Natural Language Reasoning with First-Order Logic. arXiv:2209.00840. Kirichenko, P. et al. (2025). AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions. arXiv:2506.09038. Laban, P. et al. (2023). Are You Sure? Challenging LLMs Leads to Performance Drops in The FlipFlop Experiment. arXiv:2311.08596. Lin, S., Hilton, J., Evans, O. (2022). Teaching Models to Express Their Uncertainty in Words. arXiv:2205.14334. Perez, E. et al. (2022). Discovering Language Model Behaviors with Model-Written Evaluations. arXiv:2212.09251. Sharma, M. et al. (2023). Towards Understanding Sycophancy in Language Models. arXiv:2310.13548. Tafjord, O., Dalvi Mishra, B., Clark, P. (2021). ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language. Findings of ACL. arXiv:2012.13048. Tian, K. et al. (2023). Just Ask for Calibration. EMNLP. arXiv:2305.14975. Wang, B., Yue, X., Sun, H. (2023). Can

ChatGPT Defend its Belief in Truth? Findings of EMNLP. arXiv:2305.13160. Wen, B. et al. (2024). Know Your Limits: A Survey of Abstention in Large Language Models. TACL. arXiv:2407.18418. Xiong, M. et al. (2024). Can LLMs Express Their Uncertainty? ICLR. arXiv:2306.13063. Yin, Z. et al. (2023). Do Large Language Models Know What They Don't Know? Findings of ACL. arXiv:2305.18153.

Appendix A. Limitations and Dual-Use / Ethical Considerations

Dual use. The forged checks are an attack: a message that looks like verification and changes a model's answer. Publishing them is net positive. The construction is simple (alter one line of a derivation), it is already implied by the sycophancy literature, and the benchmark gives defenders a measured, exactly scored test of whether a model verifies. The forgeries are confined to synthetic Boolean networks; nothing in them transfers as a ready attack on a deployed system.

Scope of the check. The exact checker certifies status only inside the declared class. Pre-sprint development runs (R5 Track M) showed that when the class is wrong, the checker can certify a false claim. Presenting the check as general verification would overclaim.

Model family. The benchmark was implemented by Claude (Claude Code, Opus 5.5) and directed by Claude (Fable); Claude Opus 5.5 is also evaluated. Every Claude request ran in an empty directory with no tools, no settings, no memory and no project instructions, and the request record was audited for injected context.

Thinking summaries. For Claude the stored reasoning is the route's thinking summaries, not the raw reasoning; it is reported as such and never scored.

Benchmark cost. At high reasoning effort a frontier reasoning model did not finish a Tier H turn within 32000 completion tokens. The online cells are incomplete because of that cost and of the shared subscription window, not by selection.

Human subjects and data. None; no personal data appears in any prompt. Online models were used under their providers' terms.

AI tool use. Code, runs, analysis and the first draft of this report were produced by Claude Opus 5.5 under the direction and review of Claude Fable and the author. Every number in the report is generated by `report/fill.py` from the analysis outputs; the build fails if the text holds a number that is not in them.

Appendix B. Pre-sprint and sprint work; lock, amendments and predictions

Pre-sprint (2026-11-09 to 2026-11-12, disclosed). Exact engines and SAT cross-check; the sprint benchmark (Task S, Task P, P0–P4), Tier W, derivations, the verifier and the forged checks; the R7b gates; design lock v1 (commit `1e79358`, 2026-11-12T15:06:24-06:00); the local main panel.

Sprint (2026-11-13 onward). Lock addendum v2 (commit `f8e1fb3`, 2026-11-13T09:15:16-06:00; first P5 request 2026-11-13T15:17:12Z, first online request 2026-11-13T15:18:03Z): P5, reasoning storage, online caps, predictions. Amendment v2-a1: Claude thinking summaries flag. v2-a2: DeepSeek cap 16000 → 32000 after a failed smoke (the one allowed repair). v2-a3 (commit `309fb1c`, 2026-11-13T14:35:24-06:00; first resumed Claude request 2026-11-13T20:35:49Z): the Claude priority order of blocks (run order only; items, prompts and scores unchanged), the usage-limit pause, the Saturday 12:00 CST hard stop, the DeepSeek low-effort condition, the spend stop on the upper bound. Every commit is listed in `PREEXISTING_WORK.md`.

Predictions against outcomes (12 met, 5 not met; a prediction whose cell has fewer than 12 finished worlds, or whose model was not run, is "not testable in this run"). The lock does not define "the floor" used by two predictions; post-hoc, a model is off the Tier W floor when the lower bound of its balanced accuracy exceeds the always-"not settled" row at the reported precision (llama-3.3-70b is the edge case and counts as at the floor). The rule that decides each outcome and its values are in `reports/sprint_all/report_data.json`.

lock	by	prediction	outcome
v1	director	v1-D1: On Tier W at least one local model is off the floor, and every local model still gives a definite verdict on more than 20 % of open claims.	**not met**
v1	director	v1-D2: On Tier H no local model beats the surface rule.	**met**
v1	director	v1-D3: On Tier W, check discernment is below 0.5 for every local model of 24B or less.	**met**
v1	director	v1-D4: On Tier H, adoption under P3 and under P4 differ by less than 0.15 for every local model.	**met**
v1	implementation agent	v1-A1: Tier W: check discernment (plain) is below 0.25 for every local model of 24B or less, and adoption of the forged status under P4 is at least 0.6 for each of them.	**met**
v1	implementation agent	v1-A2: Tier W: corruption under P4 is at least 0.5 for every local model of 24B or less.	**met**
v1	implementation agent	v1-A3: Tier W: no local model's balanced accuracy exceeds the Tier W surface rule (0.540 on the main panel) by more than 0.10.	**met**
v1	implementation agent	v1-A4: llama-3.3-70b, Tier W: check discernment below 0.5.	**met**
v1	implementation agent	v1-A5: gemma-4-12b: more than 25 % of its turn-1 Tier W replies have no parseable answers (token cap 6000, D83).	**met**
v2	director	P-D1: P-D1. Both online models have Tier W Task S balanced accuracy above the surface rule (0.540), interval lower bound above it.	**met**
v2	director	P-D2: P-D2. Both online models have Tier W check discernment under P4 above 0.30.	**met**
v2	director	P-D3: P-D3. Under P5f, each online model quotes the forged line exactly in more than 50 % of forged checks; each of the six local models in fewer than 15 %.	**not met**
v2	director	P-D4: P-D4. P5 lowers forged adoption relative to P4 by more than 0.15 for every model off the Tier S floor on Tier W; for the floor models the change is within +-0.15.	**not met**
v2	director	P-D5: P-D5. Tier H check discernment of both online models is below 0.15.	**not testable in this run**
v2	implementation agent	P-A1: P-A1. Every local model has P5f exact detection below 0.10 and lenient detection below 0.25 (point estimates).	**not met**
v2	implementation agent	P-A2: P-A2. Claude Opus 5.5 has P5f exact detection above 0.60 and P5v false alarm below 0.10 (point estimates).	**met**

lock	by	prediction	outcome
v2	implementation agent	P-A3: P-A3. Both online models have Tier W check discernment under P4 above every local model's, and at least one of them above 0.50 (point estimates).	**met**
v2	implementation agent	P-A4: P-A4. For at least 5 of the 6 local models, forged adoption under P5f is within 0.15 of forged adoption under P4 on Tier W (the instruction to verify alone does not move them).	**not met**
v2	implementation agent	P-A5: P-A5. At least one online model has Tier W Task S balanced accuracy above 0.80 (point estimate), and neither exceeds 0.60 on Tier H.	**not testable in this run**

Appendix C. Worked example

See `reports/R7B_WORKED_EXAMPLE.md` for a full Tier W task, its valid check, the forged check (one phantom row), the verifier's finding and the P5 message; reproduced from a pilot world never shown in the main panel.

Appendix D. Online accounting and P5 descriptors

model	tier	P0	P1	P2	P3	P4
Claude Opus 5.5	W	1/24	0/24	2/24	24/24	24/24
Claude Opus 5.5	H	0	0	0	0	0
DeepSeek V4-Pro (low)‡	W	1/24	1/24	2/24	24/24	24/24
DeepSeek V4-Pro (low)‡	H	0	0	0	0	0

Table 6. Finished / planned worlds per online cell (main panel). The Claude run had a shared subscription window and a hard stop (Saturday 12:00 CST); its blocks ran in a fixed priority order: Tier W P3 + P4, P5, Tier W P0–P2, Tier H P3 + P4, Tier H P0–P2.

condition	turns parsed (Tier W, Tier H)	turns stopped at the cap	completion tokens	pass
DeepSeek V4-Pro, effort high, cap 16 000	0/8 (0/4, 0/4)	8	16000–16000	no
DeepSeek V4-Pro, effort high, cap 32 000	2/8 (2/4, 0/4)	6	28843–32000	no
DeepSeek V4-Pro, effort low, cap 32 000	6/8 (4/4, 2/4)	2	3652–32000	yes
Claude Opus 5.5, effort high	8/8 (4/4, 4/4)	0	713–48177	yes

Table 7. Online smokes on 4 pilot worlds (8 turns). Pass rule fixed before each smoke: at least 6 of 8 turns parsed and every Tier W turn parsed.

model	requests sent (all)	of which rejected without inference	worlds finished: main / P5	recorded cost
Claude Opus 5.5	353	177	51/180 / 30/60	$15.77 (notional, subscription)
DeepSeek V4-Pro (low)‡	255	0	52/180 / 60/60	$8.90 at request-time price; $17.79 upper bound

Table 8. Online accounting. Requests rejected by the subscription window reached no model and are counted separately. Claude's cost is the CLI's notional figure on a subscription.

Two properties of the forged messages are reported, not repaired (post-hoc, D104). (i) For 1 of the 115 forged named claims the forgery alters more than one line; quoting any of them counts as a hit. Excluding it changes exact detection only for gemma-4-12b (0.106 → 0.109). (ii) Because forgeries are per claim, one message can show the same table entry with two different values, a contradiction visible without the evidence rows. Such messages are not detected more often: mistral quotes the line in 0.232 of messages with a visible contradiction against 0.203 without. The same property holds for the locked P4: gemma-3-4b adopts the forged status in 0.906 of P4 messages with a visible contradiction (10 of 24).